for the composite S_{AB} system. How could you tell whether the corresponding state is entangled? I am not referring to an experimental test but to a mathematical procedure. A related question is whether there are varying degrees of entanglement. If there are, how could you quantify them?

Entanglement is the quantum mechanical generalization of correlation. In other words, it indicates that Alice can learn something about Bob's half of the system by measuring her own. In the classical example of the previous lecture, I illustrated the idea of correlation using coins. If Alice observes the coin that Charlie gave her, she not only knows whether her own coin is a penny or a dime; she also knows which coin Bob has. That's the experimental picture. The *mathematical* indication of correlation is that the probability function $P(a, b)$ does not factorize (that is, it does not look like Eq. 6.3). Whenever the probability distribution does not factorize, there are nonzero correlations as I described in Inequality 6.2.

7.7.1 The Correlation Test for Entanglement

Let's assume that **A** is an Alice observable and **B** is a Bob observable. The correlation between them is defined in terms of the average values (also known as the expectation values) of the individual observables, and of their product. Suppose that

$$\langle \mathbf{B} \rangle$$

$$\langle \mathbf{AB} \rangle$$

are these expectation values. The correlation $C(A, B)$ between **A** and **B** is defined as

$$C(A, B) = \langle \mathbf{AB} \rangle - \langle \mathbf{A} \rangle \langle \mathbf{B} \rangle.$$